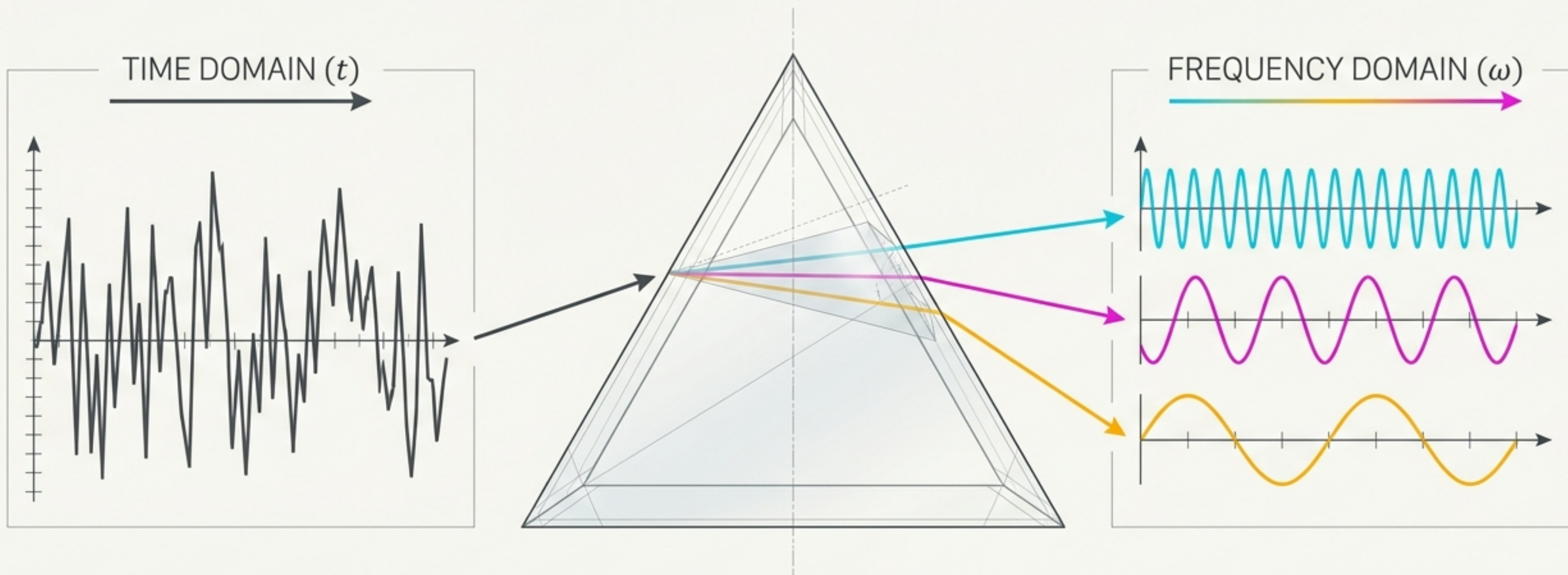
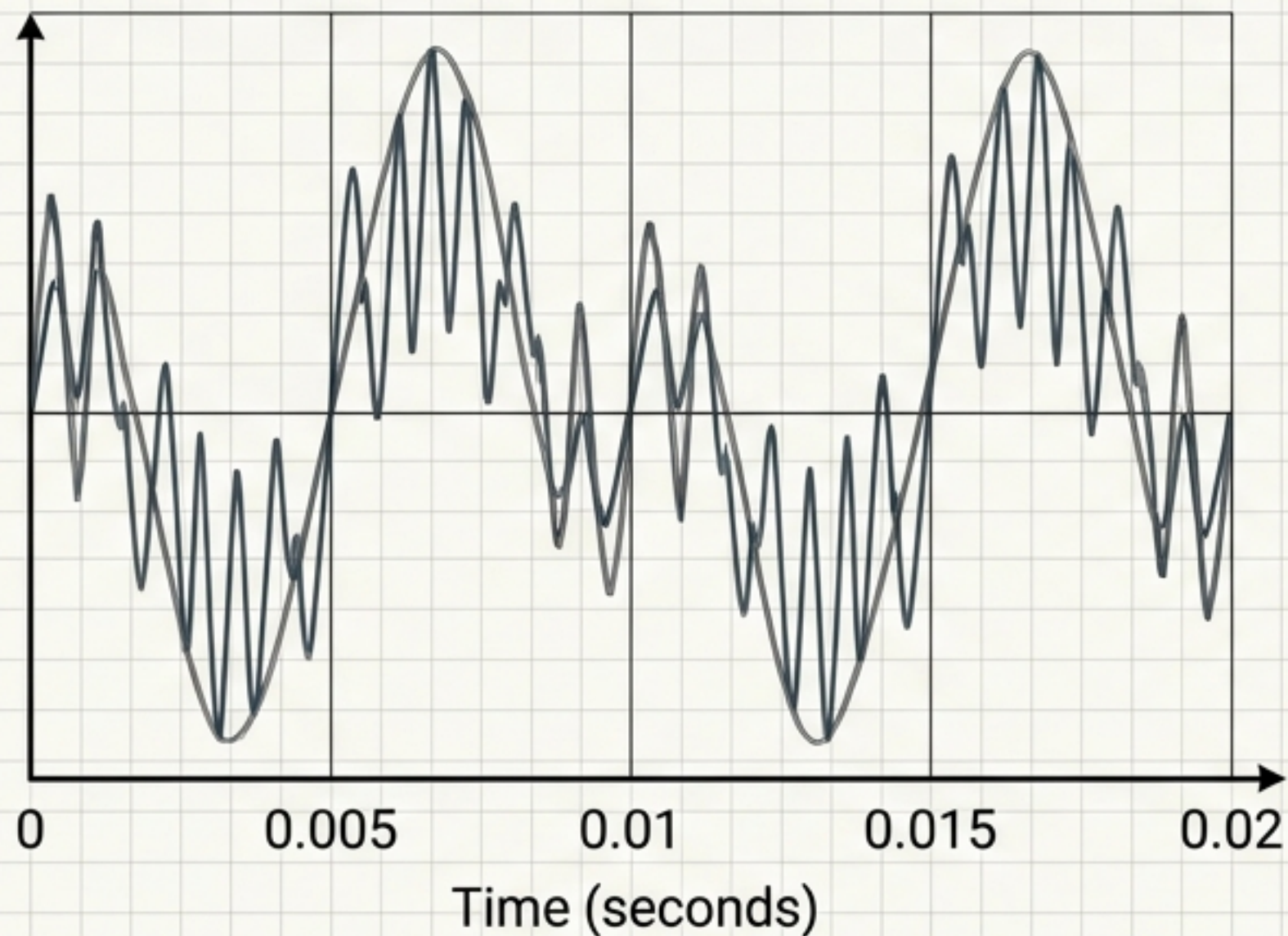




A sufficiently well-behaved signal can be represented as a combination of sinusoids or complex exponentials.

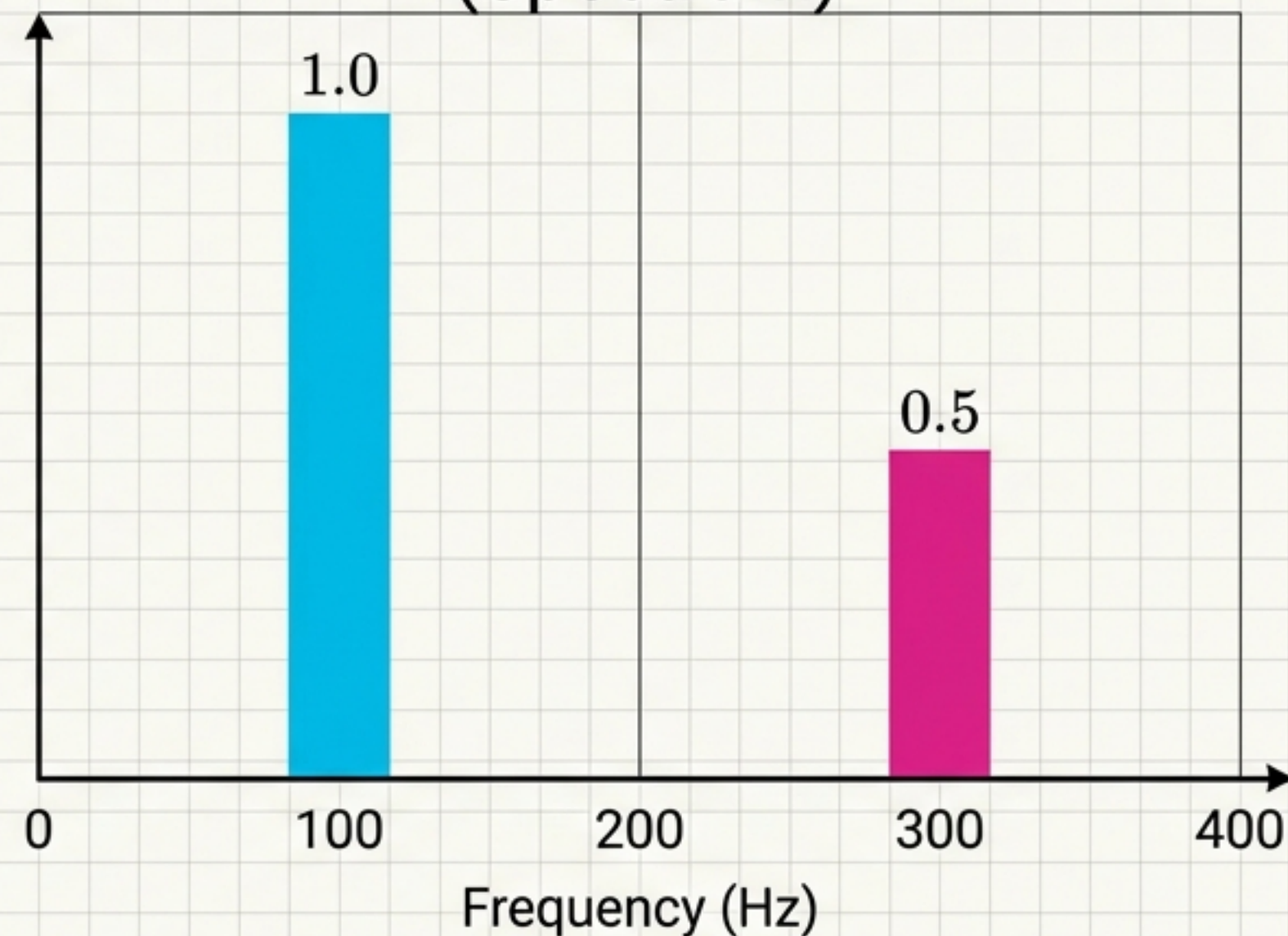
Time-Domain Representation $x(t)$



$$x(t) = \cos(2\pi 100t) + 0.5\cos(2\pi 300t)$$

Describes how amplitude changes from instant to instant.

Frequency-Domain Representation (Spectrum)



$$\omega = 2\pi f \quad f = \frac{\omega}{2\pi}$$

Unpacks the signal into discrete components, revealing the dominant 100 Hz and 300 Hz frequencies.

The Natural Basis of Fourier Analysis

Euler's formula provides the foundational bridge. Complex exponentials are not mere mathematical notation; they inherently capture both the magnitude and phase required to completely specify a signal component.

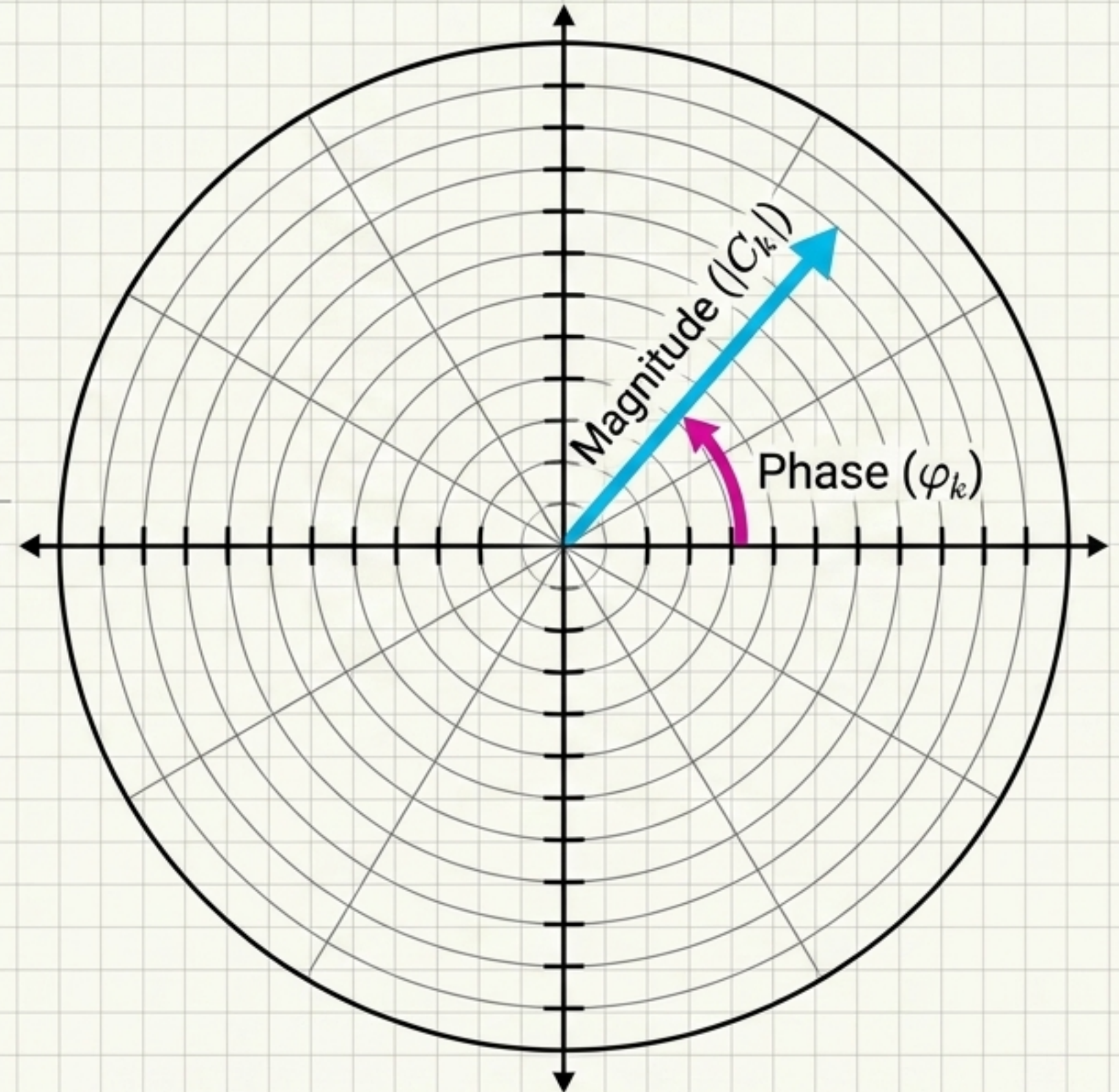
Euler's Formula:

$$e^{j\theta} = \cos\theta + j\sin\theta$$

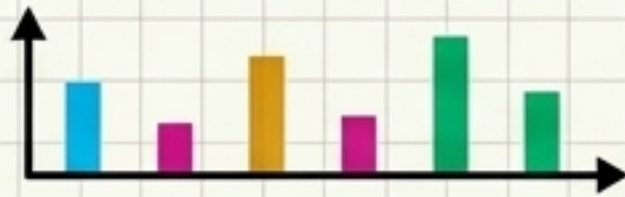
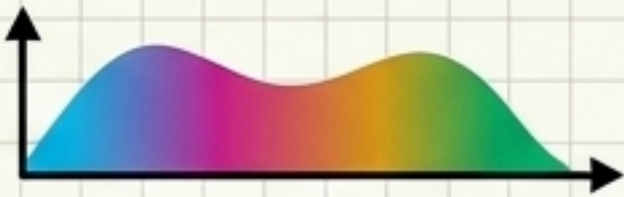
Complex Signal Representation:

$$X(f) = |X(f)|e^{j\angle X(f)}$$

Phase/Magnitude Compass



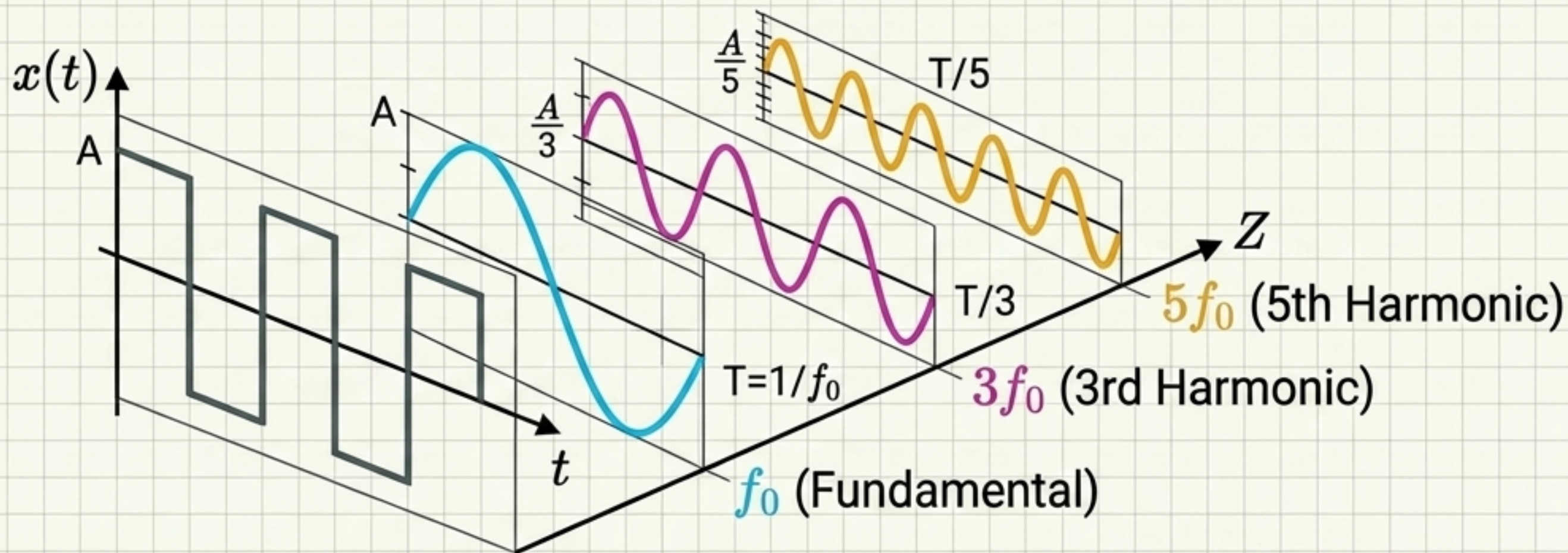
Choosing the Right Analytical Tool

	Fourier Series	Fourier Transform
Signal Type	Periodic Signals (loops forever)	Aperiodic Signals (finite duration, transients)
Frequency Variable	Discrete Harmonics ($k\omega_0$)	Continuous Frequency (f)
Math Operation	Summation (Σ)	Integral ($\int$)
Output Shape		
Output Notation	C_k	$X(f)$

Rule of Thumb: Periodic signals break down into discrete steps.
Aperiodic signals smear across a continuous spectrum.

Deconstructing a Periodic Waveform

The Anatomy of a Harmonic

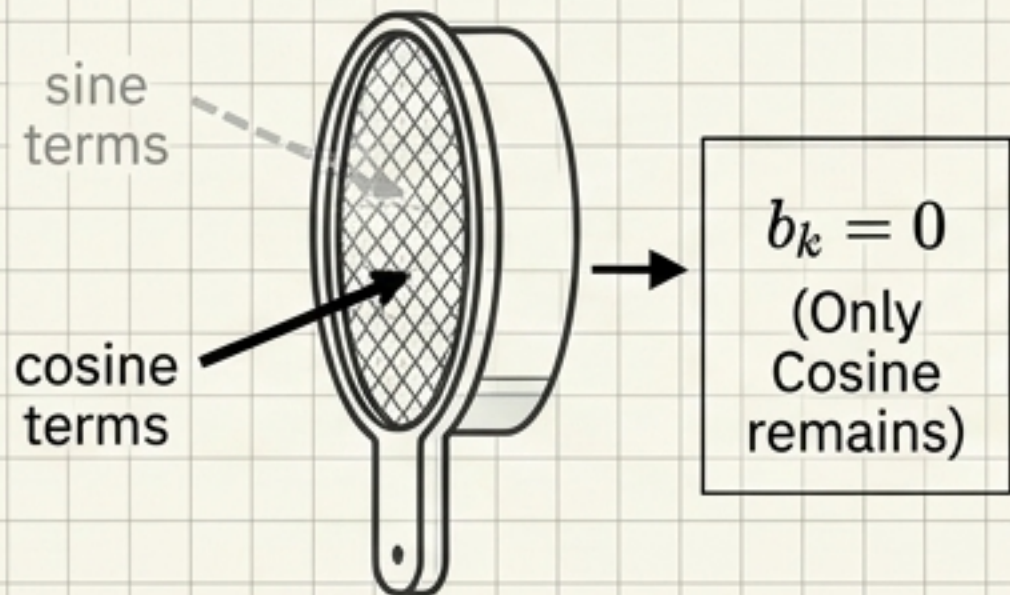
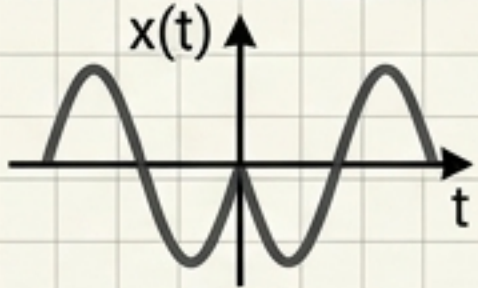
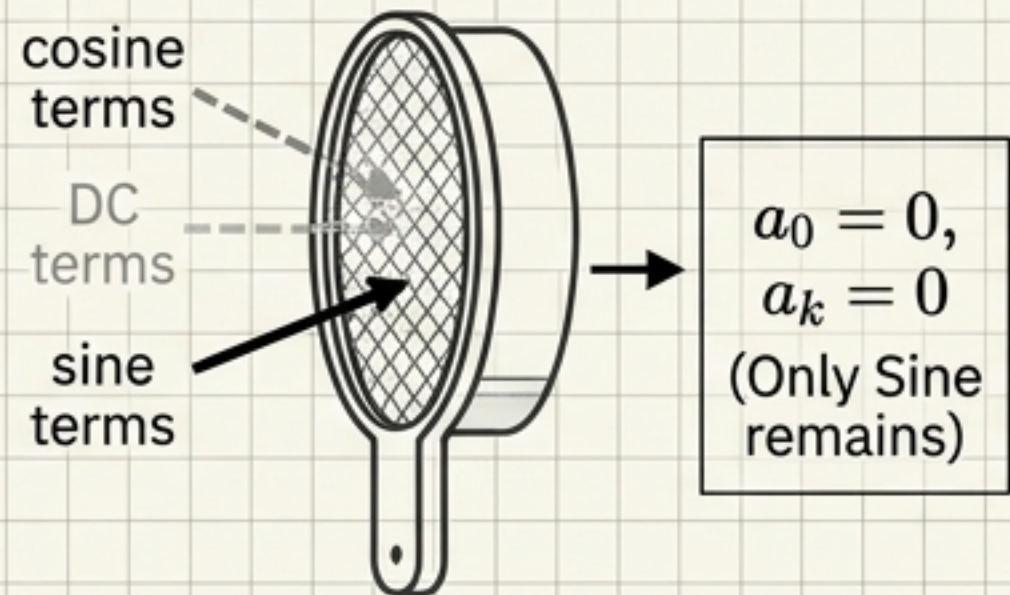
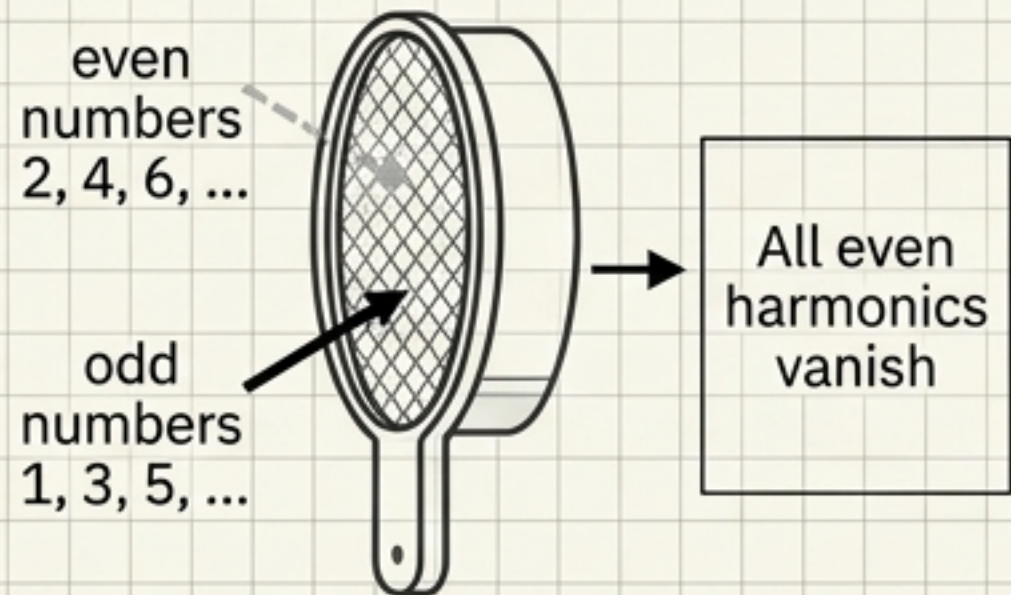


Harmonic Frequency = $k \times$ Fundamental Frequency

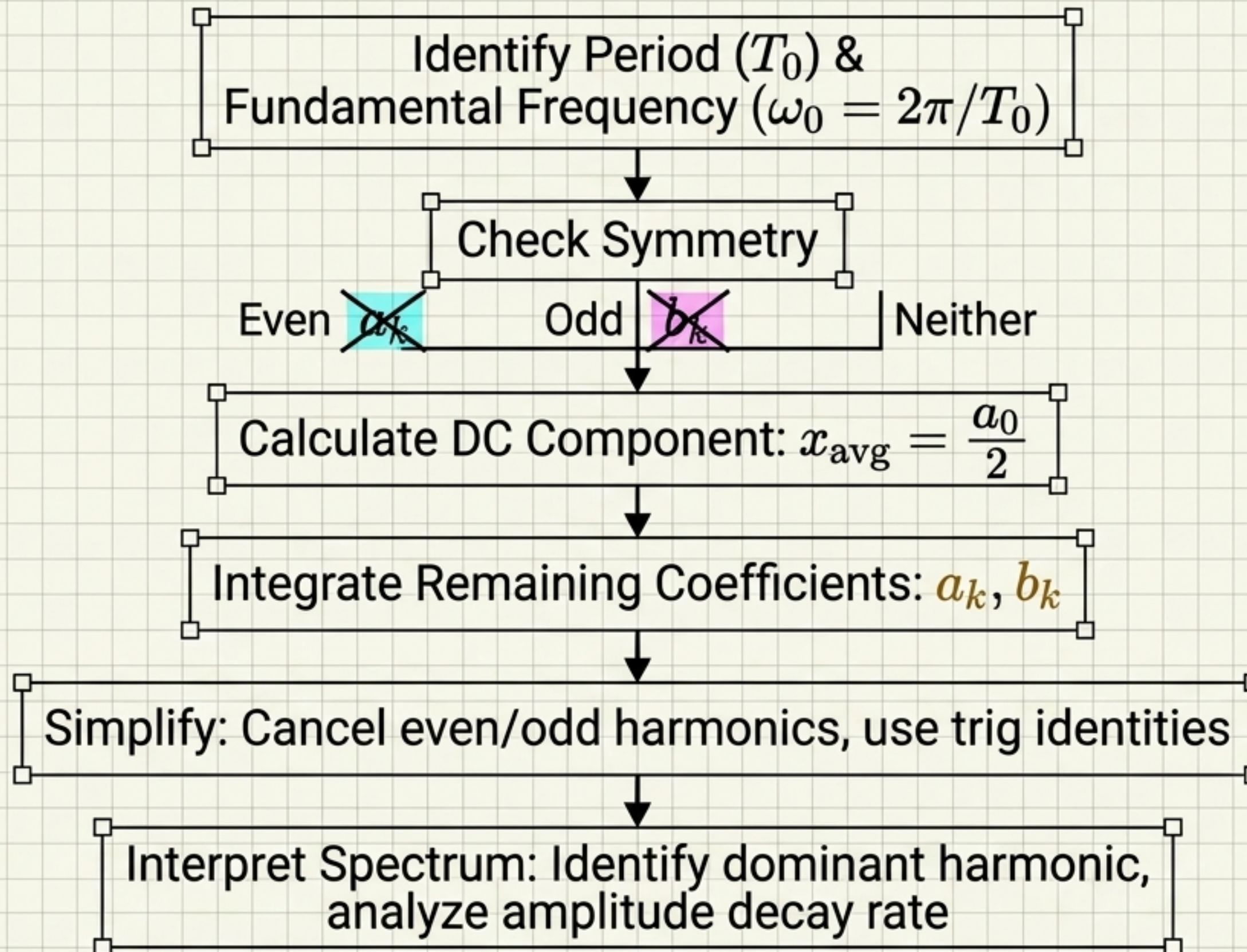
$$x(t) = \frac{4A}{\pi} \left[\sin(\omega_0 t) + \frac{1}{3} \sin(3\omega_0 t) + \frac{1}{5} \sin(5\omega_0 t) + \dots \right]$$

⚠ **Gibbs Phenomenon:** Reconstructing sharp discontinuities with truncated harmonics results in oscillatory overshoot.

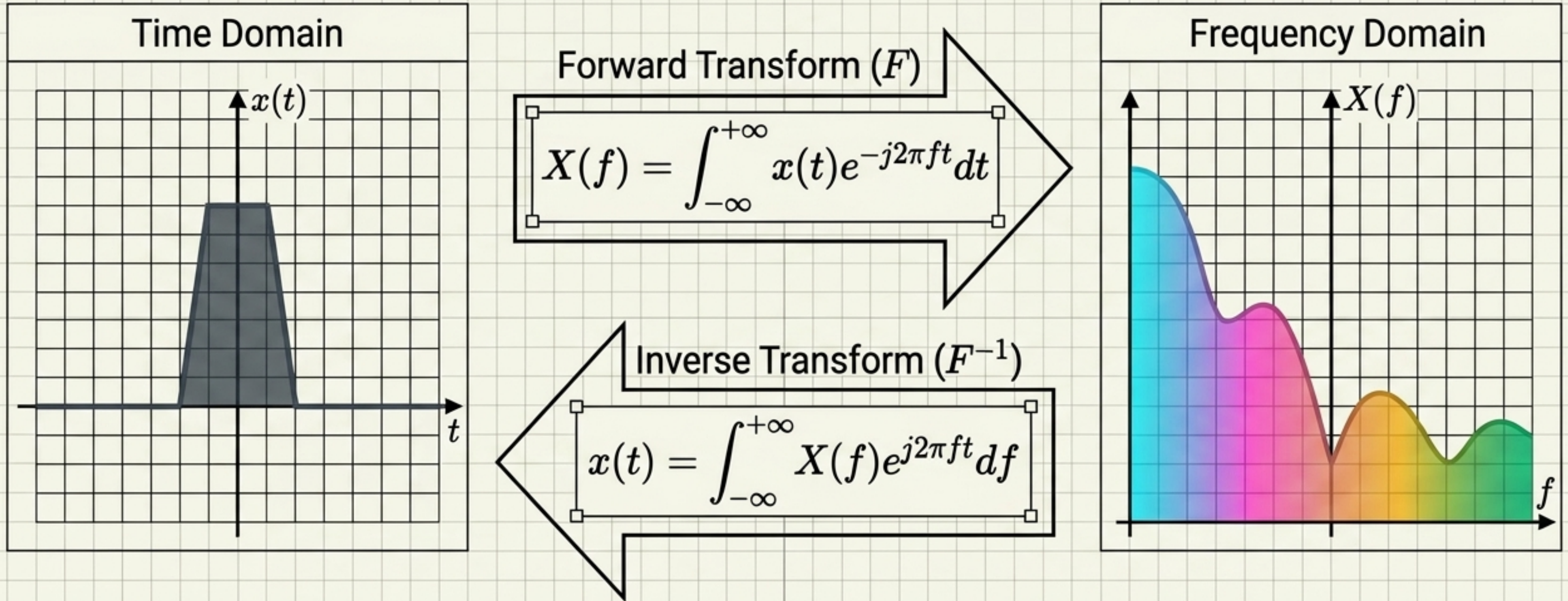
Signal Symmetry as a Calculation Bypass

Symmetry Filter Matrix		
1 Even Symmetry	2 Odd Symmetry	3 Half-Wave Symmetry
 $x(-t) = x(t)$  $b_k = 0$ (Only Cosine remains)	 $x(-t) = -x(t)$  $a_0 = 0,$ $a_k = 0$ (Only Sine remains)	 $x\left(t + \frac{T_0}{2}\right) = -x(t)$  All even harmonics vanish
Before calculating any integral, inspect the waveform for symmetry. It can eliminate up to half of your mathematical workload.		

The Periodic Signal Playbook



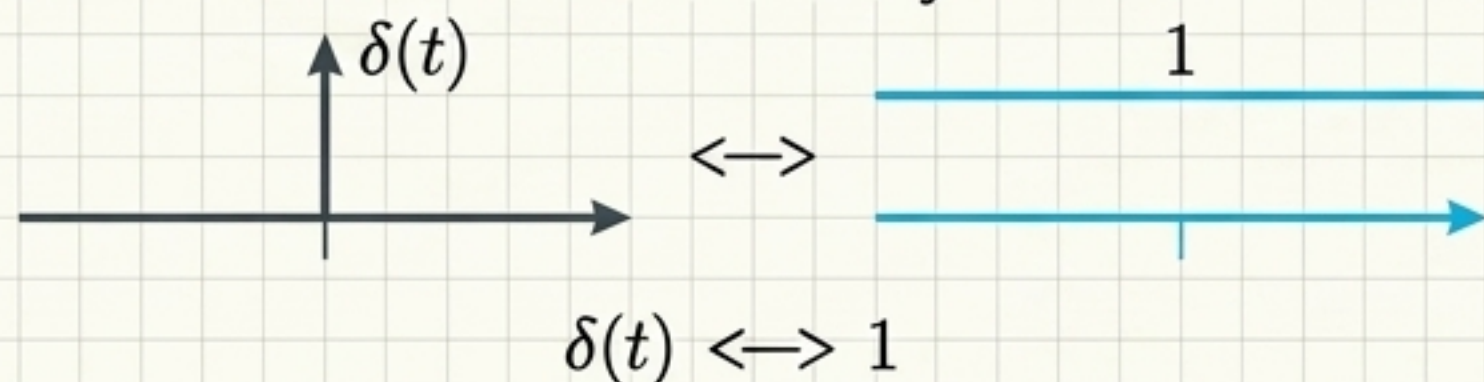
Mapping Aperiodic Signals to Continuous Spectra



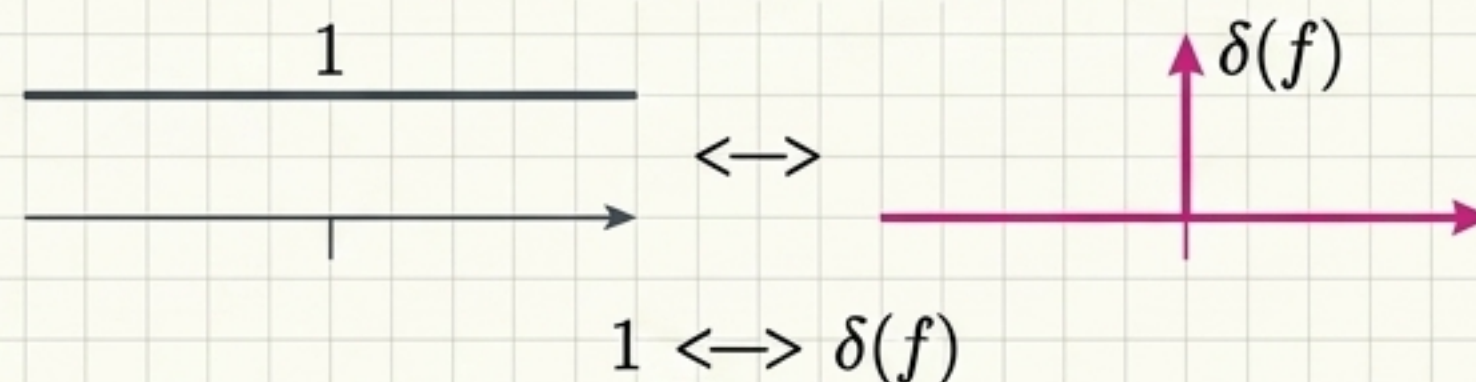
Conceptually, the Fourier Transform is the limiting form of Fourier-series analysis as the period expands to infinity. The discrete harmonics merge into a continuous frequency variable.

The Vocabulary of Transforms

The Duality



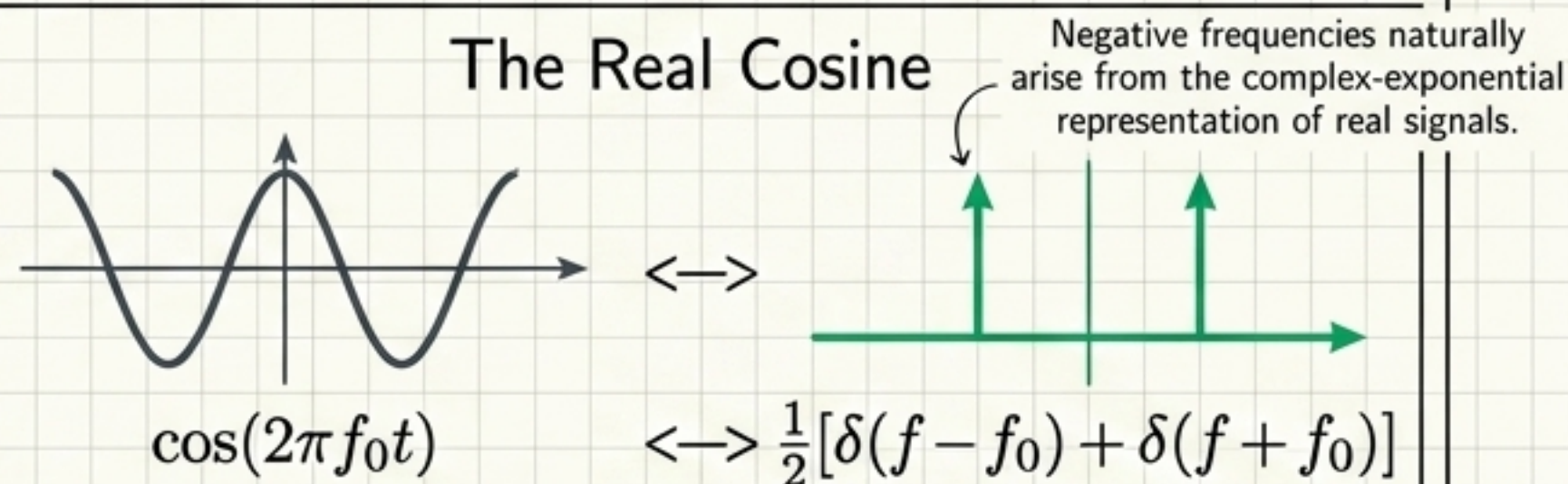
The Inversion



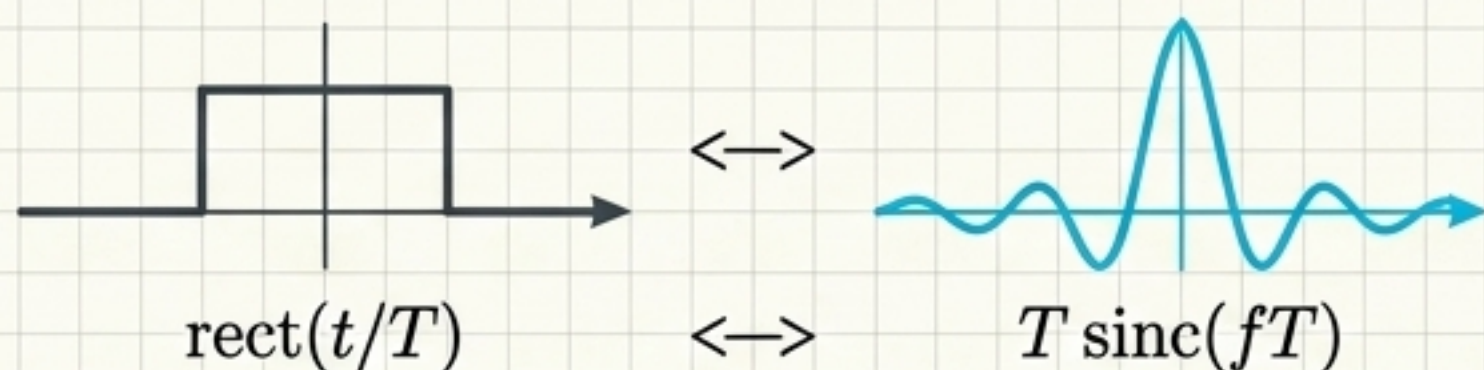
The Exponentials



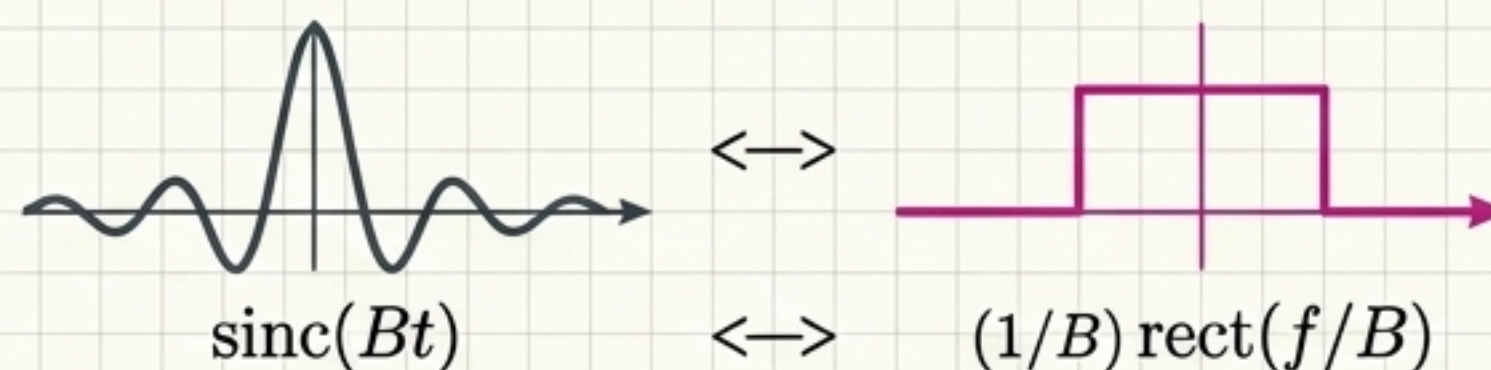
The Real Cosine



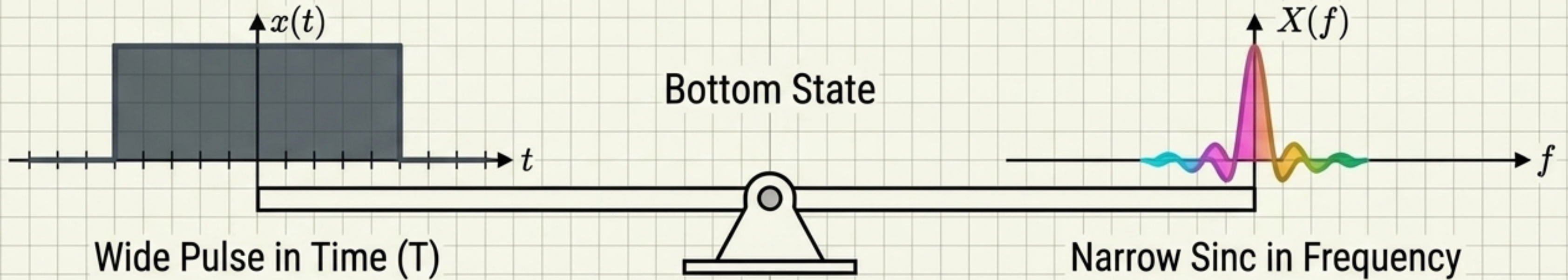
The Pulse



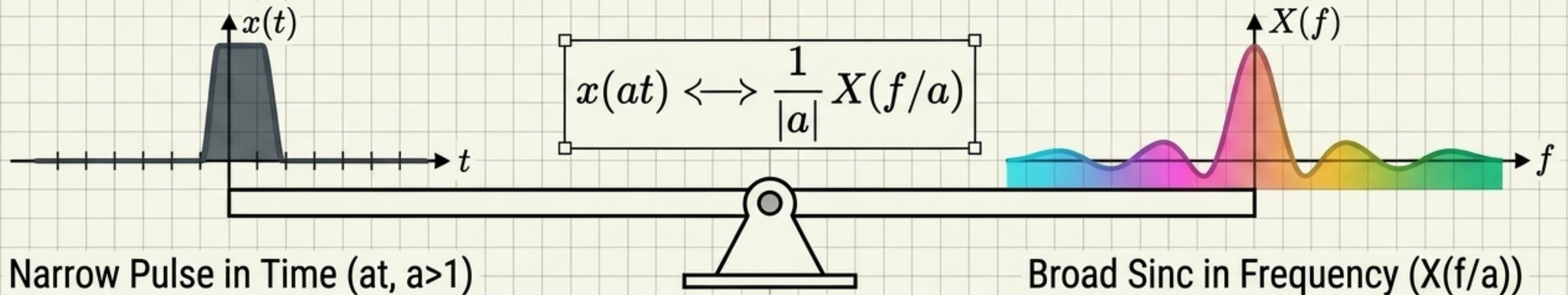
The Sinc



The Time-Frequency Seesaw



Shorter in time $\iff$ Broader in frequency

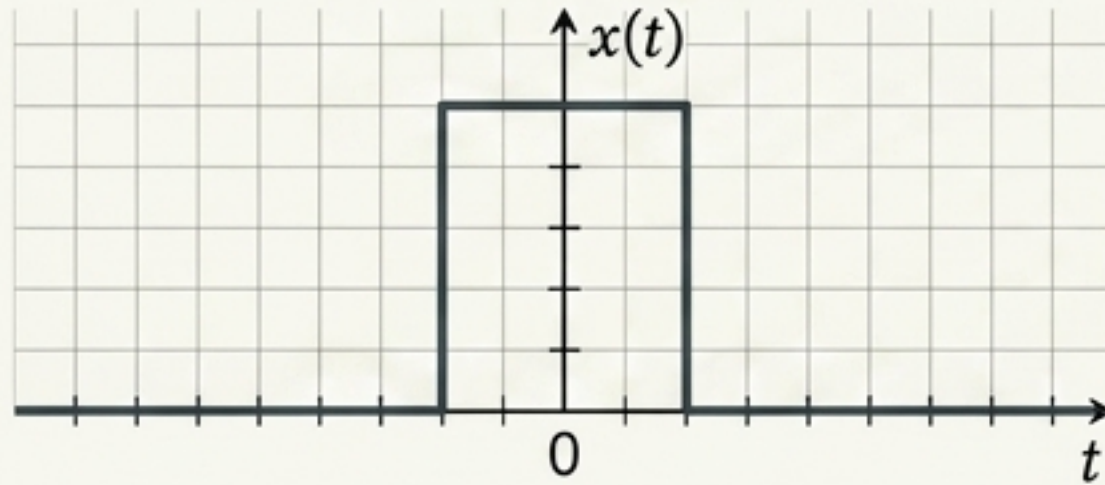


For a pulse of width T , the main-lobe bandwidth is $\Delta f = 2/T$. Decreasing T forces Δf to increase.

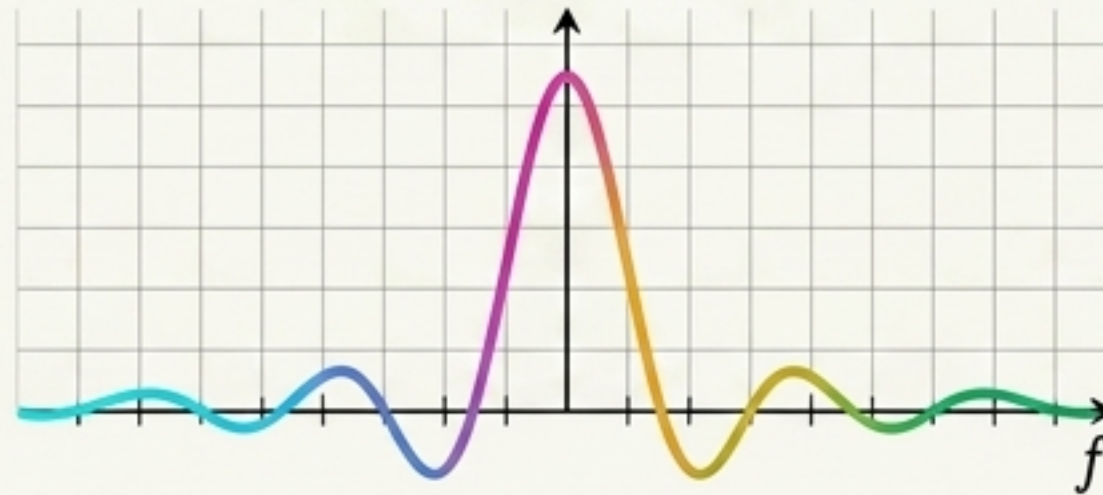
The Phase Shifter

Unshifted

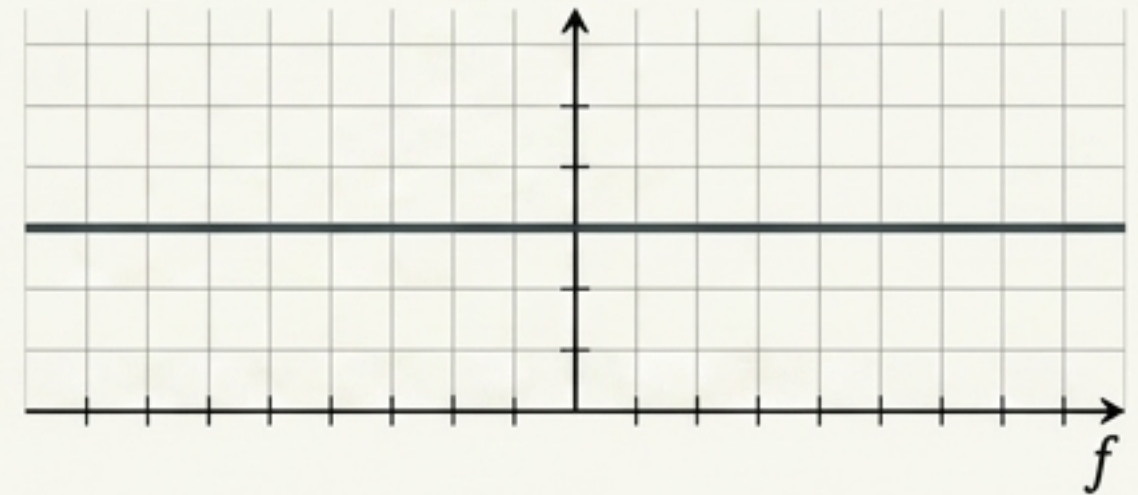
Time Domain: $x(t)$



Magnitude Spectrum: $|X(f)|$

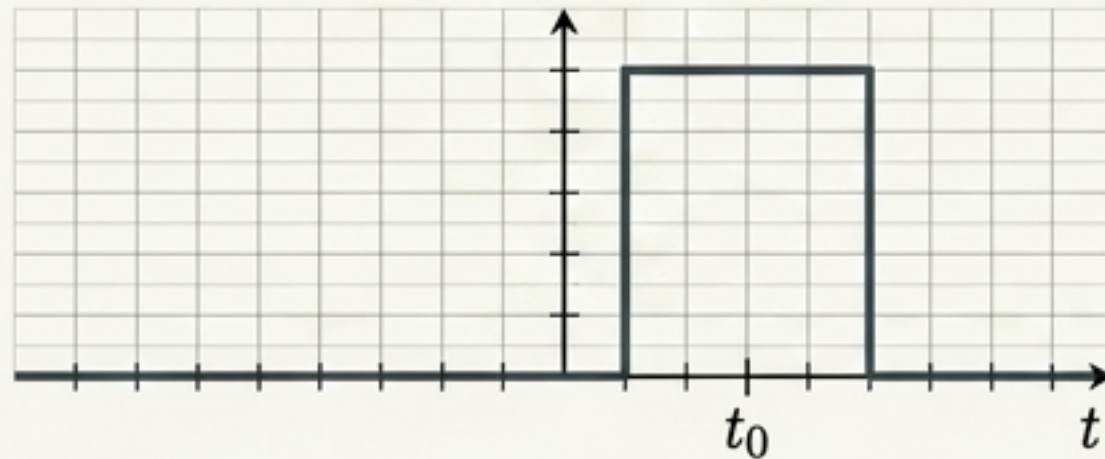


Phase Spectrum: $\angle X(f)$

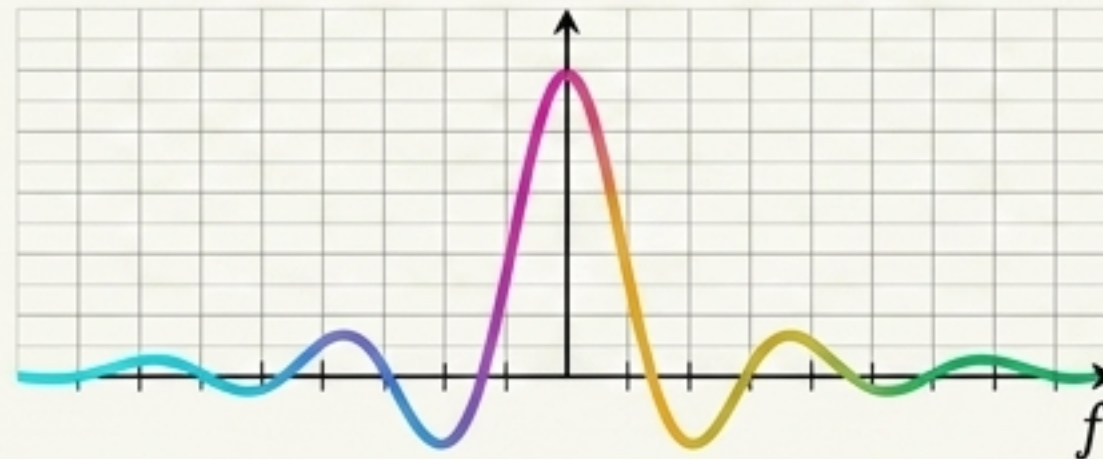


Shifted

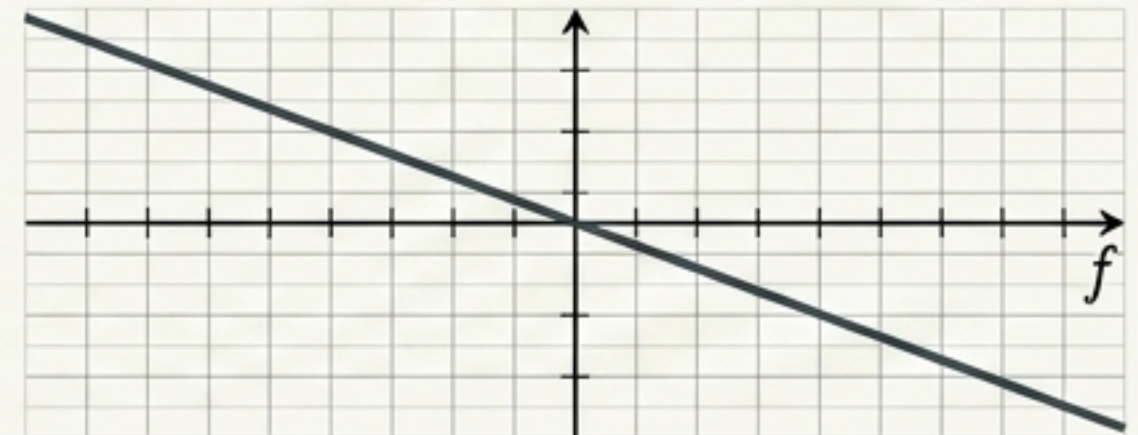
Time Domain: $x(t-t_0)$



Magnitude Spectrum: $|X_{\text{shifted}}(f)|$



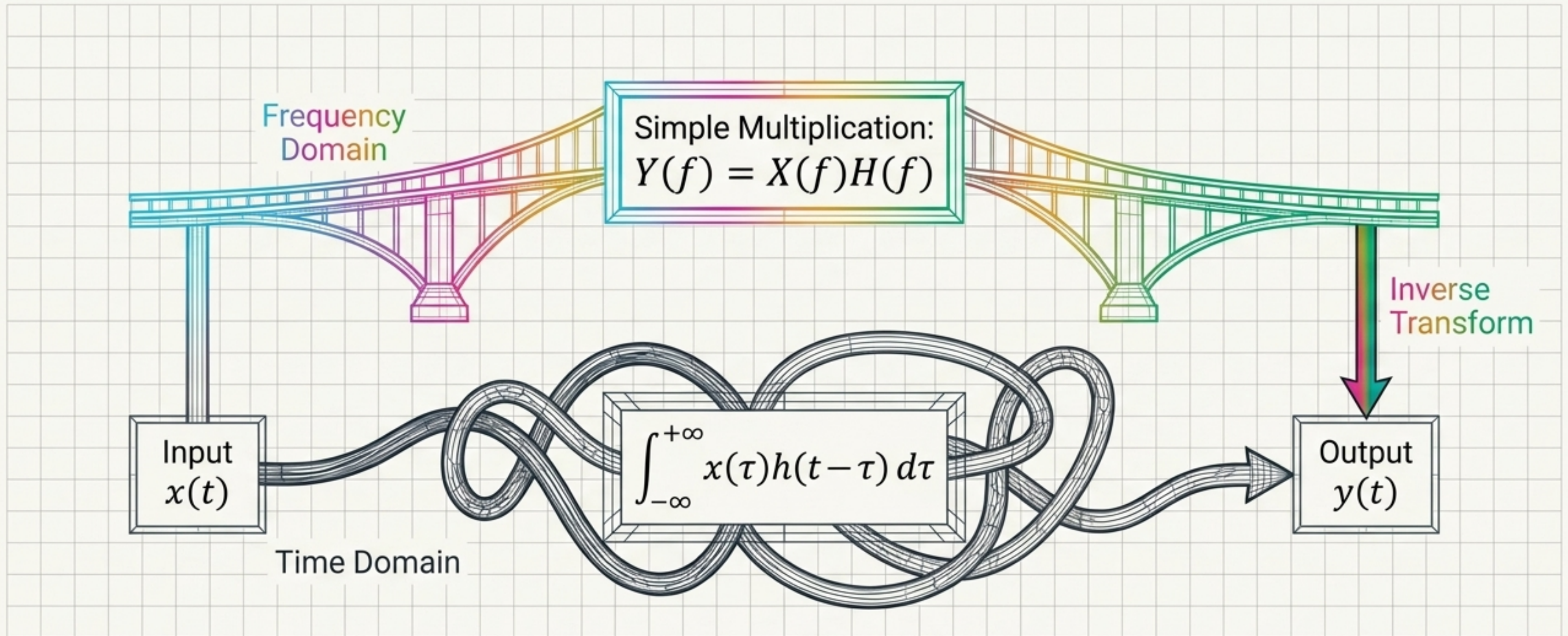
Phase Spectrum: $\angle X_{\text{shifted}}(f)$



$$x(t - t_0) \longleftrightarrow X(f)e^{-j2\pi ft_0}$$

$|X_{\text{shifted}}(f)| = |X(f)|$. A delay in time rotates the spectral phase, but the distribution of **frequency** strength remains identical.

The Convolution Shortcut

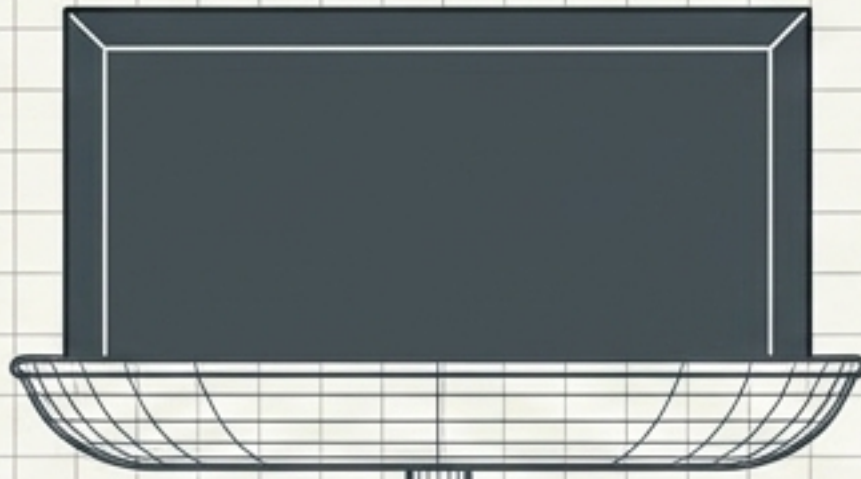


Convolution in time becomes multiplication in frequency. This theorem is the mathematical foundation that makes frequency-domain filtering possible.

Tracking Energy and Power

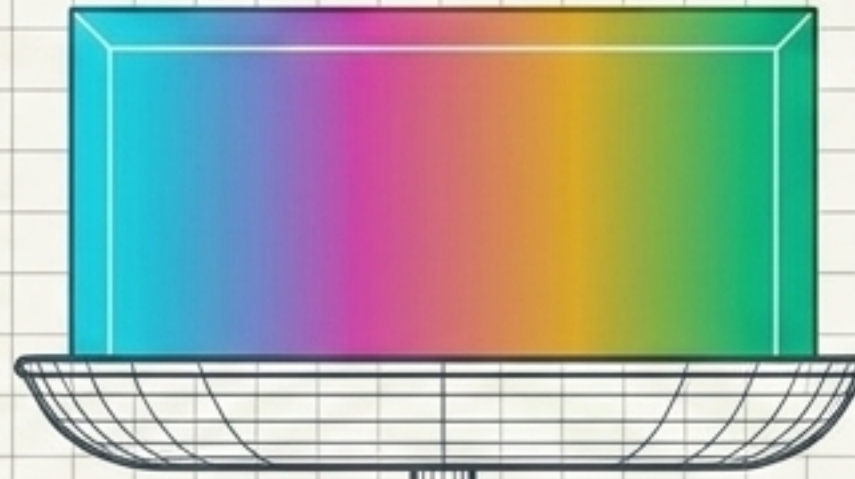
Total Signal Energy:

$$\int |x(t)|^2 dt$$



Total Spectral Energy:

$$\int |X(f)|^2 df$$



Power Spectral Density (PSD) is fundamental for analyzing:

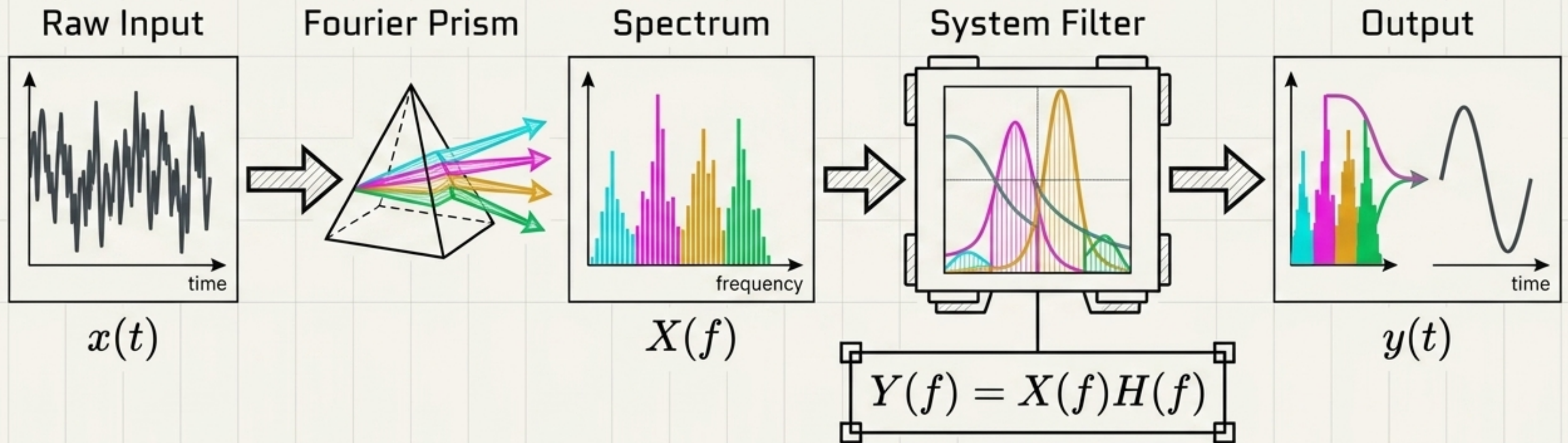
- Noise
- Communications
- Control Systems
- Instrumentation

Energy Spectral Density:

$$S_E(f) = |X(f)|^2$$

Parseval's Theorem: Total average power in the time domain equals the total power distributed among the Fourier components.

The Signal Processing Assembly Line



This chain justifies the entire Fourier framework. By decomposing a signal into its spectral DNA, we can precisely manipulate, filter, and control real-world systems.